



DREAMNET

A beginner's guide to building the internet



3. More resources

I was first inspired to research technology like this by Lori Emerson's work with Other Networks <https://othernetworks.net/>

I've already mentioned the fantastic hydroponic trash blog which inspired this zine and which gave me the blueprint for the network <https://blog.hydroponictrash.solar/offgridinternet/>

You can even repurpose a disposable vape and start running a server <https://bogdanthegeek.github.io/blog/projects/vapeserver/>

Iffy books have some incredible instructional intro zines <https://iffybooks.net/zines/>

Mastodon is a great social network to start going down the rabbit hole for things like this - you can sign up and start following hashtags like #indieweb #solarpunk #lowtech and more!

Low tech web

As we're running your site on a raspberry pi from your own home and not a fancy cloud server, you'll want to consider the size of your website files (things like images and audio) and the processing power needed (things like javascript).

Googling 'low tech web design' will give you some great resources to make your site eco and pi friendly. Things like using plain HTML/CSS/JS or dithering images are a great way to start.



0. Intro

What is this?

A brief starter guide to learning about network infrastructure and servers, a spring board for diving deeper into the world of protocols, servers, ports and routers and such.

Who is this for?

If you are curious about the idea of building your own server, setting up a wifi router, or building networks of any kind, this is a nice intro to create a proof-of-concept that you could build on. If you are at all nervous about any of the words used already, please be reassured that that's totally natural! There's a lot of jargon and opaque language around these tools, but there's also lots of amazing resources out there to learn.

That said, I won't be explaining every step in detail - I'm assuming some knowledge of things like using command line interfaces, how to edit files in the terminal, and a basic mental model of a raspberry pi/installing software via ssh. If you do get stuck don't panic! Making mistakes and breaking things is a great way to learn, and the answer is probably a short Google or youtube video away. I'm also happy to answer any questions - you can contact me on harriethw at gmail dot com or on mastodon at <https://mas.to/@harriethw>

Why?

Aside from it just being fun to learn and build things, I do think that it is important to at least imagine a world where our infrastructure is not built by Big Tech. With the climate emergency, rise of fascism, and even just the expense of technology, reclaiming the networks we use to communicate on could be a useful tool for communities.

There are quite a few examples now from communities that have built communication networks themselves that we can take inspiration from:

<https://www.tucsonmesh.net/> - offering high speed internet access to their community, built by and maintained by the community

<https://communitytechnology.github.io/docs/cck/> - documentation from groups working to provide community networks in Detroit and elsewhere.

<https://www.apc.org/en/guifinet> - Guifi.net is an independent initiative in Spain

Handy guide to ports

Ports have unique numbers and they are a bit like doors for your server, entering through the right door tells the server which corridor to take you down...

80 - the default port used by web servers. Your browser will use this port to try and talk to your router.
443 - the default port for secure connections to servers, so it's used by https
21 - the default FTP port

You can change these default ports via dietpi as an option for added security (e.g. a bot won't be able to target your FTP because it won't know what port its on).

i ***What the hell is headless?***

never used the command line
before? Don't panic!

Command line interfaces (CLI's) can be a bit intimidating - most people only see them on screens when someone is an elite hacker - but there's nothing mysterious about them.

Think of it as a way to talk to your computer, but instead of using your voice, mouse or touchscreen to tell it what to do, you're using magical word spells.

Getting to grips with the CLI is super fun and unlocks a whole host of powerful and speedy tools at your fingertips.

There's a CLI cheatsheet at the end of this booklet to remind you of some commands you might use regularly. *(TBA April 2023)*

If you've never done something like this before, you may want to check out Free Code Camp or another resource on using the command line or bash scripting for beginners.

<https://www.freecodecamp.org/news/command-line-for-beginners/>

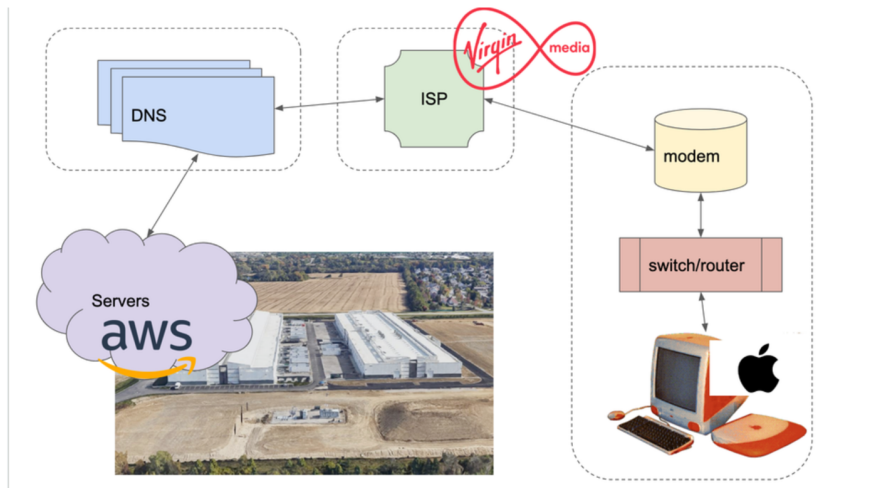
Find this online at

https://github.com/Harriethw/dreamnet/blob/main/documentation/how_to.md



1. What is the internet??

The internet as we know it now is a complex and sprawling system, built over decades, with janky layers of protocols, software, hardware and shifting power dynamics. If you were to slim it down to the bear basics though, it might look something like this



The next steps involve a lot of text and copy and pasting, so it's best to switch over to the digital version of these instructions! They basically explain the technical steps of how to setup your server and router.

As a bonus material for this zine - I've included some crib sheets on some of the technical terms and tools needed in the next few pages, as well as some further reading :)



This is heavily cribbed from the fantastic Hydroponic Trash newsletter, with some changes that I needed to get it working or felt simplified the process. Check out their blog here for more amazing resource, and how to make the network we are about to build solar powered or scaled to a whole city service!

<https://blog.hydroponictrash.solar/offgridinternet/>

In this setup - your Raspberry Pi is going to act like a **server**. It will host a website using a minimal-coding blog library called Hugo. It could also host other things like

- A free library of PDFs
- Photos and videos
- An instant messaging server

We'll use a regular travel or home **modem/router** to act both as a modem and router, as well as to provide a **DNS** and a Captive Portal functionality.

This means that when our users connect to our wifi, they won't be connected to the wider internet but instead our own little ecosystem on the Raspberry Pi server. It could be described as a **Local Area Network (LAN)**

You could scale this by having many servers attached to the same network, as well as many antennas/nodes/routers to boost the initial signal of the network.

1. What is the internet??

A **Server** is a computer with some data or content on it that is open to the public on an IP address (which looks something like 198.51.100.1)

A **Domain Name Server (DNS)** is another computer that resolved human readable addresses like www.example.com to those IP addresses of the servers

A **Modem** provides your connection to your Internet Service Provider, and wider internet traffic that the Router then directs.

A **Router** is like the traffic conductor of data flowing through the internet, your home one connects your local network to the wider internet, and service providers have larger and more complicated ones.

When you make a call to a website from a device in your home, it goes through a router, modem and then is directed to the web server, which can then send a response back. Simple!



Indigenous Anarchist Federation - <https://iaffaio.org.wordpress.com/2020/10/11/skills-for-revolutionary-survival-5-communications-equipment-for-rebels/>



If you want to go further down the rabbit hole, there's also tons of non-internet based communication networks like HAM Radio, LoRA, Meshtastic etc. etc.

2. Build your own network

We don't need global infrastructure to make something like this happen in our home or community. There are some great libraries to look at like LoRa, Meshtastic and Reticulum, but for this example I've used a plain old travel router with Wifi and a web server running on a Raspberry Pi.

